

SRMS-005 — Structural Recognition as Runtime Intelligence

Structural Recognition above Metric Similarity (SRMS)

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Repository: <https://github.com/sizhet/Structural-Recognition-above-Metric-Similarity>

Abstract

Structural recognition is often viewed as a classification problem.

This view is incomplete.

Recognition does not merely assign labels.

Recognition activates computation.

Once a structure has been recognized, a new runtime becomes available.

Different Calling Graphs, Runtime Invariants, constraints, certified components, execution paths, and decision rules become applicable.

This paper proposes a new interpretation:

Structural Recognition is Runtime Activation.

Recognition is therefore not simply the first stage of perception.

It is the entry point of runtime intelligence.

1. Recognition Is More Than Classification

Many AI systems treat recognition as the assignment of a category.

For example:

Image

↓

Cat

or

Sentence

↓

Question

Such classification is useful.

However, engineering systems require much more.

After recognition, the system must know:

- what operations are allowed,

- what invariants must hold,
- which Calling Graph should execute,
- which Runtime Components become available,
- which safety rules apply,
- which decision boundaries exist.

Recognition therefore changes the runtime itself.

2. Recognition Activates Runtime

Consider a software runtime.

Before recognition:

Unknown Object

After recognition:

Database Transaction

Immediately,

new runtime capabilities become available:

- transaction manager,
- rollback,
- commit,
- isolation rules,
- concurrency control,
- logging,
- recovery.

Nothing changed physically.

Only recognition changed.

Yet an entirely different runtime has become active.

Recognition is therefore runtime activation.

3. Runtime Depends on Structural Identity

The same observation appears everywhere.

Examples include:

Medical

Recognize:

Emergency Patient

Runtime changes to:

- emergency protocol,
- critical monitoring,
- rapid diagnosis,
- priority scheduling.

Aviation

Recognize:

Emergency Landing

Runtime changes to:

- emergency procedures,
- dedicated Calling Graph,
- different safety constraints,
- different decision priorities.

AI Coding

Recognize:

JUnit Test

Runtime changes to:

- testing APIs,
- assertions,
- execution environment,
- validation workflow.

Recognition determines which runtime becomes active.

4. Recognition Determines Runtime Invariants

Every runtime contains invariants.

Examples include:

- required states,
- legal transitions,
- resource ownership,
- synchronization,
- safety constraints,
- interface compatibility.

These invariants cannot be determined by similarity.

They depend entirely on structural recognition.

Recognition therefore identifies not only an object.

It identifies the invariant system surrounding that object.

5. Runtime Activation Through Calling Graphs

Calling Graphs provide a natural interpretation of runtime activation.

Recognition determines:

Which Calling Graph should execute?

Examples:

Normal Mode

activates

Normal Calling Graph

while

Emergency Mode

activates

Emergency Calling Graph

The recognition step switches the computational pathway.

6. Recognition and Runtime Invariants

Runtime Invariants describe structural identities that remain preserved across implementations.

Recognition discovers:

- invariant identity,
- invariant constraints,
- invariant interfaces,
- invariant runtime behavior.

Recognition therefore serves as the gateway to Runtime Invariants.
Without recognition,
Runtime Invariants cannot be selected.

7. Recognition Before Decision

Decision making should therefore follow:

Input



Recognition



Runtime Activation



Runtime Invariants



Decision
rather than

Input



Similarity



Decision

The former respects runtime structure.

The latter risks comparing objects that belong to different computational worlds.

8. Recognition and Certified Components

Recognition also determines which certified components become available.

For example,

recognizing:

Sorting Task

may activate:

- certified sorting algorithms,

- verified APIs,
- benchmark datasets,
- performance constraints.

Recognizing

Scheduling Task

activates a completely different component ecosystem.

Recognition therefore controls certified runtime reuse.

9. Recognition Is a Runtime Switch

Recognition behaves like a switch.

Before recognition:

Unknown Runtime

After recognition:

Known Runtime

The runtime now exposes:

- APIs,
- Calling Graphs,
- Runtime Invariants,
- Certified Components,
- Constraints,
- Decision Rules.

Recognition changes the computational universe.

10. Runtime Intelligence

We therefore define Runtime Intelligence as:

The capability to activate the correct computational runtime through structural recognition.

This capability includes:

- recognizing structural identity,

- selecting the appropriate runtime,
- activating Runtime Invariants,
- selecting Calling Graphs,
- identifying certified components,
- respecting decision boundaries,
- supporting reliable execution.

Recognition is therefore not merely perception.
Recognition is computation.

11. Relation to Structural Recognition above Metric Similarity

The previous SRMS papers established that recognition must precede similarity.

This paper explains why.

Similarity measures proximity.

Recognition activates runtime.

These are fundamentally different computational functions.

Similarity compares objects.

Recognition determines which computational world those objects belong to.

12. Toward Runtime-Centric AI

Future AI systems should increasingly separate two responsibilities.

Structural Recognition

Determines:

- identity,
- runtime,
- invariants,
- Calling Graph,
- certified components.

Metric Similarity

Determines:

- proximity,
- preference,
- ranking,
- retrieval.

This separation produces AI systems that are more interpretable, modular, and reliable.

Key Takeaways

- Recognition is more than classification.
- Recognition activates runtime.
- Runtime determines Calling Graphs, Runtime Invariants, APIs, and constraints.
- Similarity cannot activate runtime.
- Runtime intelligence begins with structural recognition.
- Future AI architectures should be runtime-centric rather than similarity-centric.

SRMS Principle 005

Structural Recognition is Runtime Activation. Recognition determines which computational runtime becomes available. Similarity may rank objects within that runtime, but it cannot establish the runtime itself.